Sepsis Signal

AI-powered early warning for sepsis in the ICU

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THE PROBLEM

Sepsis is detected too late in the ICU.

Early warning is difficult when critical signals are incomplete, constantly changing, and easy to misinterpret.

01 — FRAGMENTED DATA

Vital signs and lab results are irregular and often missing.

02 — MISSED SIGNALS

Deterioration unfolds over time, across multiple clinical indicators.

03 — FALSE ALARMS

More sensitive alerts can mean more false positives and alert fatigue.

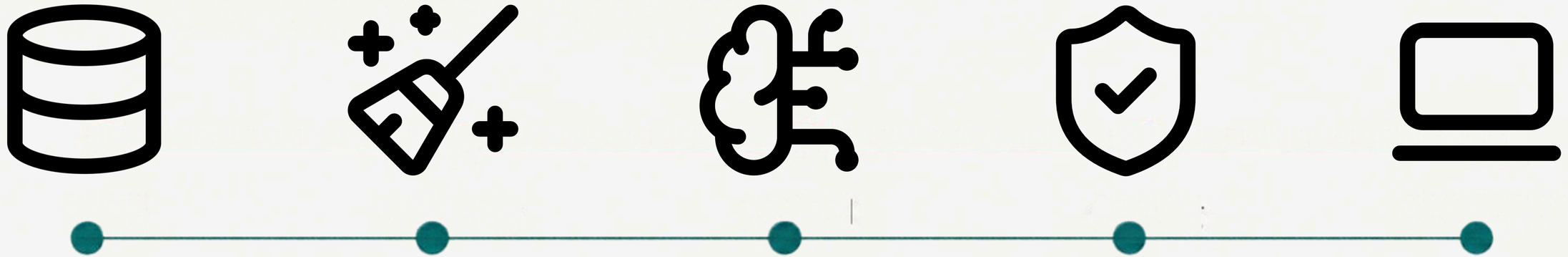


WHY IT'S HARD

- **Time:** risk changes hour by hour, so the model only sees history up to the prediction point
- **Missing data:** an unmeasured lab isn't a normal lab, so we flag it and track time since last measurement
- **Imbalance:** positive hours are rare, so we judge on AUPRC, precision and recall, not accuracy
- **Leakage**: patient-level splits, training-only fitting, no filling from the future
- **Alert fatigue:** catching more cases means more false alarms, and we show that trade-off

HOW IT WORKS

From signals to early insight.



We turn ICU data into an early warning signal to support faster, more informed decisions.

OUR STACK

Data	PhysioNet 2019 (CC BY 4.0), local pandas & NumPy
Modeling	scikit-learn baseline, XGBoost/LightGBM, optional PyTorch LSTM/GRU
API	FastAPI serving model and evaluation results
Dashboard	Next.js + React + TypeScript with interactive charts
Dev tools	Claude & coding assistants, Canva for deck

Everything runs locally. No patient data goes to external AI services.

Raw data → pandas / scikit-learn → XGBoost / PyTorch → FastAPI → Next.js dashboard

What we built.

01

A leakage-resistant pipeline: patient-level splits, no future data, thresholds chosen on validation only.

02

Time-aware features: latest values, rolling mean/min/max, trends, time since last measurement, missingness flags.

03

An honest comparison of a simple baseline against a sequence model on the same held-out patients.

Model overview.

Model	Approach	Features
Logistic Regression	Simple baseline	Engineered time-aware features
XGBoost / LightGBM	Tree-based	Same features
LSTM / GRU (stretch)	Sequence model	Last 6–24 hours

COMPETITORS (“WHERE WE FIT”)

Tool	What it is	Where it stands
Epic Sepsis Model	Built into the Epic EHR, already used by hospitals without FDA review	A 2021 study found it performed worse than expected
Prenosis Sepsis ImmunoScore	EHR-integrated ML software that uses 22 parameters to produce a risk score and four risk categories; it is not an alert system	FDA de novo authorization, the first for an AI sepsis tool (April 204)
Academic and startup tools (e.g. Johns Hopkins, Bayesian)	Similar tools from Johns Hopkins and Epic, with success reported as spotty	Mostly research or limited deployment
SepsisSignal	Research prototype on a public benchmark	Not clinically validated, by design

Business model

WHO IT IS FOR	ICU researchers, clinical informatics teams, and students evaluating early-warning models before any clinical use
THE PROBLEM WE SOLVE	early-warning model results are hard to trust when missing data, leakage and false alarms are hidden
HOW WE MAKE MONEY	research grants and pilot partnerships first, then subscription access to the evaluation dashboard for institutions
MAIN COSTS	data infrastructure, model development, clinical advisors, and privacy, security and regulatory review
HOW WE REACH THEM	research networks, the PhysioNet community, university labs, conferences, clinical informatics groups

"RESEARCH FIRST. CLINICAL USE ONLY AFTER EXTERNAL AND PROSPECTIVE VALIDATION."

IS IT FEASIBLE?

Data	PhysioNet 2019 (40k patients, public CC BY 4.0)
Technology	Standard Python stack; baseline first, LSTM/GRU stretch
Team	3 developers: pipeline, models/evaluation, dashboard
Main Risk	Sequence model delays; baseline acts as guaranteed fallback
Path to Clinical Use	External validation, silent testing, and FDA review (out of scope)

RESULTS AND HONEST LIMITS

What we've learned so far.

RESULTS

Retrospective public data, may not match any real hospital. Does not prove prediction before clinical symptoms.

External and prospective silent-mode validation, clinician feedback, privacy, security and regulatory review.

LIMITATIONS & NEXT STEPS

- Limited to retrospective ICU data.
- Needs validation across more sites.

- Expand prospective validation.
- Integrate into clinical workflow.